

Darshit Joshi

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SUMMARY

Senior Software Engineer with strong expertise in machine learning, statistics, and Large Language Models, building end-to-end ML/GenAI pipelines from data preparation through model training, fine-tuning, evaluation, and deployment. GCP certified ML Professional Engineer skilled in Python-based solution development, translating complex business problems into scalable analytical and AI-driven systems.

SKILLS

- **Machine Learning & Statistics:** Regression, Time-series Forecasting, Classification, XGBoost, Decision Trees, Hyperparameter Tuning, Statistical Analysis, Model Evaluation
- **GenAI & LLMs:** LangChain, LangGraph, RAG, Prompt Engineering, LLM Fine-tuning, Model Evaluation & Monitoring, VectorDB, PyTorch, LoRA finetuning, Agentic AI Workflows
- **Python Development:** Python, Pandas, FastAPI, Django, SQLAlchemy, JavaScript, TypeScript, Redis, RabbitMQ, Celery, WebSockets
- **Data & Query Tools:** SQL (Query Optimization, Complex Joins, DW/BI concepts), PostgreSQL, MySQL, BigQuery, Spark, Hive
- **Deployment and DevOps:** Kubernetes, Jenkins, Docker, ArgoCD, Git, Agile, Grafana
- **AWS:** EC2, S3, ECS Fargate, Bedrock, Sagemaker, Lambda, SQS, CloudWatch
- **GCP:** BigQuery, VertexAI, DataFlow, KubeFlow

PROJECTS

- **CTR Prediction Model:** Built an end-to-end ML pipeline using XGBoost to predict ad click-through rate, covering data preparation, training, evaluation, and automated MLOps deployment.
- **GenAI Decision-Support Agent:** Designed an agentic LLM system (Bedrock) with RAG over a retrieval knowledge base (OpenSearch) to automate scheduling and decision workflows, including evaluation of response accuracy.
- **Generative Modeling (Stable Diffusion):** Hands-on project applying generative AI and model fine-tuning techniques (LoRAs, VAEs, CUDA, checkpoints) for image generation workflows.
- **Consistent Character Generator:** Text-to-image and image-to-image ML system using SageMaker, ComfyUI, and Qwen models; applied LoRAs and model quantization to optimize inference performance.
- **RAG-based Support Chatbot:** Built a Retrieval-Augmented Generation chatbot (Python, Twilio) with automated issue escalation, tracking, and follow-up, including model output quality evaluation.
- **AI Recommendation Engine:** Recommendation system built on open-source GenAI APIs to generate personalized suggestions from user data.
- **Fraud Detection Rule Optimization:** Applied statistical optimization (Optuna) to narrow trillions of permutations to the optimal combination of detector scores, maximizing fraud capture while minimizing false positives.

PROFESSIONAL EXPERIENCE

Armakuni

SR. SOFTWARE ENGINEER

Mar. 2025 - Present

- **GenAI Agentic Decision Support:** Designed and deployed an end-to-end agentic LLM system integrating **Amazon Bedrock** for reasoning and **RAG** over a knowledge base (**OpenSearch**), covering solution design, evaluation, and deployment for scheduling and availability-lookup automation.
- **RAG-based Analytical Chatbot:** Built a Retrieval-Augmented Generation chatbot in **Python** using **Twilio**, defining model evaluation metrics to measure response accuracy and reliability, with issue escalation, tracking, and follow-up capabilities.
- **ML Model Pipeline:** Built and evaluated a text-to-image/image-to-image generation pipeline using **SageMaker**, **ComfyUI**, and **Qwen models**, applying model optimization and fine-tuning techniques to improve throughput and output quality.

DosePack (Meditab)

SOFTWARE ENGINEER

Dec. 2022 - Mar. 2025

- **API Development:** Designed and implemented RESTful APIs to streamline business workflows, reduce response times by 50%, and enhance system interoperability.
- **Migration Automation:** Automated database migration processes, reducing deployment time by 40% with version control across environments. Created and published PyPi library for the same.
- **Data Engineering and Analytics:** Built **SQL-based ETL pipelines** with complex joins and aggregations for data pre-processing, feature engineering, and validation to support statistical analysis; produced daily performance metrics and reports for stakeholders.
- **Real-time Systems:** Developed real-time solutions using **WebSockets**, enabling bi-directional communication for features like **live updates**, **dynamic dashboards**, and **notification services**.
- **Authentication Systems:** Designed and implemented secure authentication systems using token-based mechanisms (**JWT**) and role-based access control (**RBAC**). Developed a seamless single sign-on (**SSO**) solution with **OAuth2.0 scopes**.
- **Database Engineering:** Enhanced backend performance by designing a well-optimized SQL database, significantly improving storage utilization and system efficiency.
- **Algorithm Optimization:** Optimized computationally intensive algorithms for **pack-filling slot customizations** and workflow optimizations.
- **Centralized pharmacy traffic manager:** Designed and implemented middleware service to manage optimized filling with set of rules for multiple US based pharmacy and filling services to improve their efficiency.

ACHIEVEMENTS

- Training and Placement Coordinator at DDU (2023).
- Interviewer for campus placement drives and senior roles at Aramakuni.
- Coordinator at DDU NSS club activities.

EDUCATION

Dharmsinh Desai University

B.TECH

CGPA: 8.72/10

Nadiad, Gujarat

July. 2019 - May. 2023